

Software Requirements Specification (SRS)

RaithuMitra – Farmer Assistance Mobile Application

1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements of RaithuMitra, a mobile application designed to support farmers with intelligent tools, agricultural services, communication, and market insights.

1.2 Scope

RaithuMitra provides:

- AI-based crop disease detection
- Weather prediction & alerts
- Crop price prediction & market trends
- Government scheme guidance
- Crop roadmap & cultivation guidance
- Farmer social networking
- E-commerce for seeds, fertilizers, machinery
- Drone spraying & labour hiring services
- Feedback & rating system
- Smart agriculture recommendation support

Platform: Android

Technology: Flutter + Firebase + APIs

2. Overall Description

2.1 Product Perspective

RaithuMitra is a standalone mobile application integrating with:

- Weather APIs
- Market price APIs
- Government data portals
- Firebase backend
- AI/ML trained models

2.2 User Classes

- Farmers (Primary users)
- Service Providers (Drone operators, equipment rental, workers)
- Administrators

2.3 Operating Environment

- Android 8 +
- Internet connectivity required for most features

2.4 Constraints

- Dependent on API responses
- Network connectivity issues in rural areas
- Prediction accuracy depends on AI data quality

3. Functional Requirements

3.1 User Registration & Profile

- Mobile/Email registration
- Farmer profile: location, land details, crop preferences

3.2 AI Crop Disease Detection

- Upload crop image / enter description
- AI analyzes and detects disease
- Provides treatment solutions
- Suggests fertilizers/pesticides

3.3 Weather Module

- Real-time weather data
- 7-day forecast
- Rainfall & climate alerts

3.4 Crop Price Prediction

- Daily crop market price display
- Trend analytics
- Market comparison

3.5 Government Schemes

- Available schemes list
- Eligibility details
- Application guidance

3.6 Crop Roadmap Guidance

- Step-by-step cultivation guide
- Fertilizer schedule
- Organic farming suggestions

3.7 Farmer Social Network

- Post queries
- Share experiences
- Messaging / Discussion

3.8 E-Commerce

- Seeds purchase
- Fertilizers
- Machinery renting
- Secure payment

3.9 Agricultural Services

- Drone spraying booking
- Labour hiring
- Service provider listing

3.10 Feedback System

- Ratings
- Service/product feedback

4. Non-Functional Requirements

4.1 Performance

- App loads within 3 seconds
- AI results within 10 seconds

4.2 Security

- Firebase authentication
- Secure data storage
- Payment security

4.3 Usability

- Simple UI
- Regional language support (future enhancement)

4.4 Reliability

- 99% uptime
- Regular backups

5. System Constraints

- Flutter framework
- Firebase backend
- REST APIs integration

6. Future Enhancements

- Voice assistant
- Multi-language
- Analytics dashboard
- Offline mode for limited features

7. Conclusion

RaithuMitra is designed to empower farmers through intelligent technology, market awareness, community support, and agricultural services ensuring productivity and sustainability.